



Name: _____

Class: _____

Date: _____

Cell biology

Time:

Marks: **59 marks**

Comments:

1.

Ragwort is a weed that grows on farmland.

Ragwort is poisonous to horses.

- (a) Plan an investigation to estimate the size of a population of ragwort growing in a rectangular field on a farm.

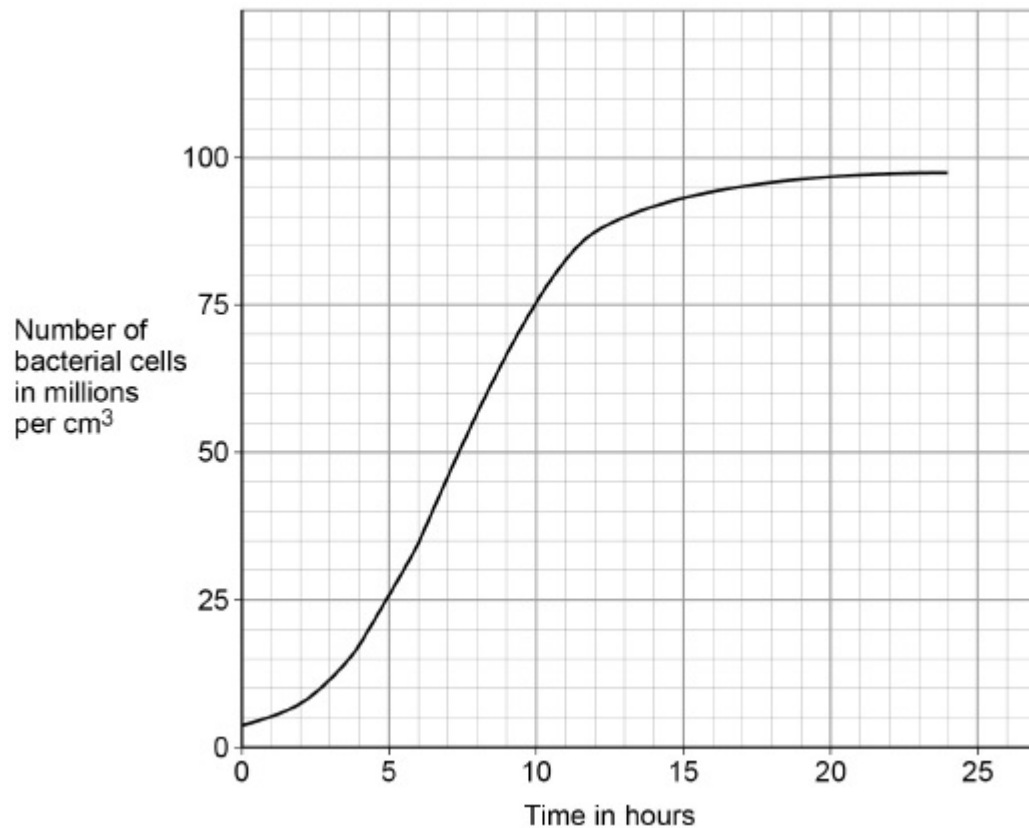
(4)

The herbicide glyphosate will kill ragwort and other weeds.

Scientists use bacteria for the genetic engineering of crop plants to make the crops resistant to glyphosate.

Figure 1 shows the growth of a culture of the bacteria in a solution of nutrients at 25 °C

Figure 1



(b) Why did the rate of reproduction increase between 2 hours and 7 hours?

(1)

- (c) After 12 hours, the rate of reproduction decreased.

Suggest **three** ways the scientists could maintain a high rate of reproduction in the bacterial culture.

1 _____

2 _____

3 _____

(3)

- (d) The rate of reproduction of the bacteria is fastest at 7 hours.

How many times faster is the rate of reproduction at 7 hours than the rate at 12 hours?

Rate at 7 hours is _____ times faster.

(4)

- (e) Scientists transferred a gene for resistance to the herbicide glyphosate into the bacteria.

The genetically-modified (GM) bacteria can then transfer the glyphosate-resistance gene to a crop plant.

Explain the advantage of making crop plants resistant to glyphosate.

(3)

(Total 15 marks)

2.

Explain how the human circulatory system is adapted to:

- supply oxygen to the tissues
- remove waste products from tissues.

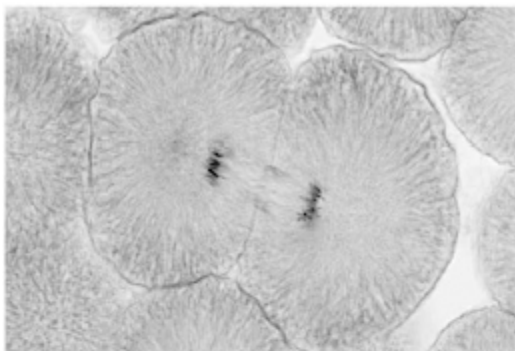
(Total 6 marks)

3.

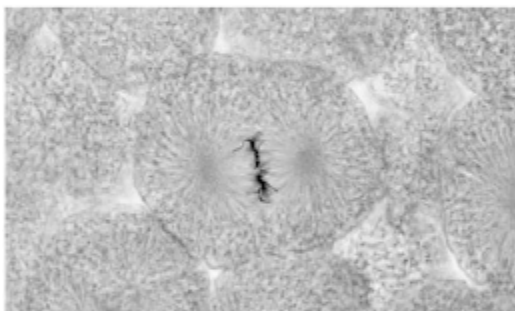
Figure 1 shows photographs of some animal cells at different stages during the cell cycle.

Figure 1

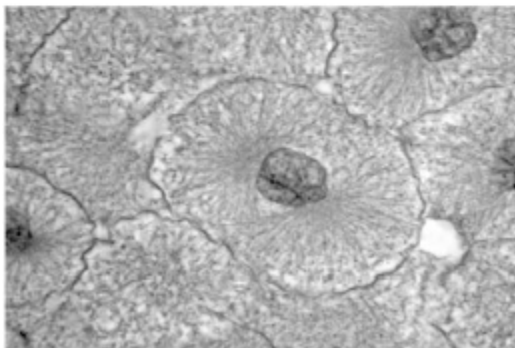
A



B



C



A © Ed Reschke/Photolibrary/Getty Images
B © Ed Reschke/Oxford Scientific/Getty Images
C © Ed Reschke/Photolibrary/Getty Images

(a) Which photograph in **Figure 1** shows a cell that is **not** going through mitosis?

Tick **one** box.

A	☐	B	☐	C	☐
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(1)

(b) Describe what is happening in photograph A.

(2)

(c) A student wanted to find out more about the cell cycle.

The student made a slide of an onion root tip.

She counted the number of cells in each stage of the cell cycle in one field of view.

The table below shows the results.

		Stages in the cell cycle				Total
	Non-dividing cells	Stage 1	Stage 2	Stage 3	Stage 4	
Number of cells	20	9	4	2	1	36

Each stage of the cell cycle takes a different amount of time.

Which stage is the fastest in the cell cycle?

Give a reason for your answer.

Stage _____

Reason _____

(2)

- (d) The cell cycle in an onion root tip cell takes 16 hours.

Calculate the length of time **Stage 2** lasts in a typical cell.

Give your answer to 2 significant figures.

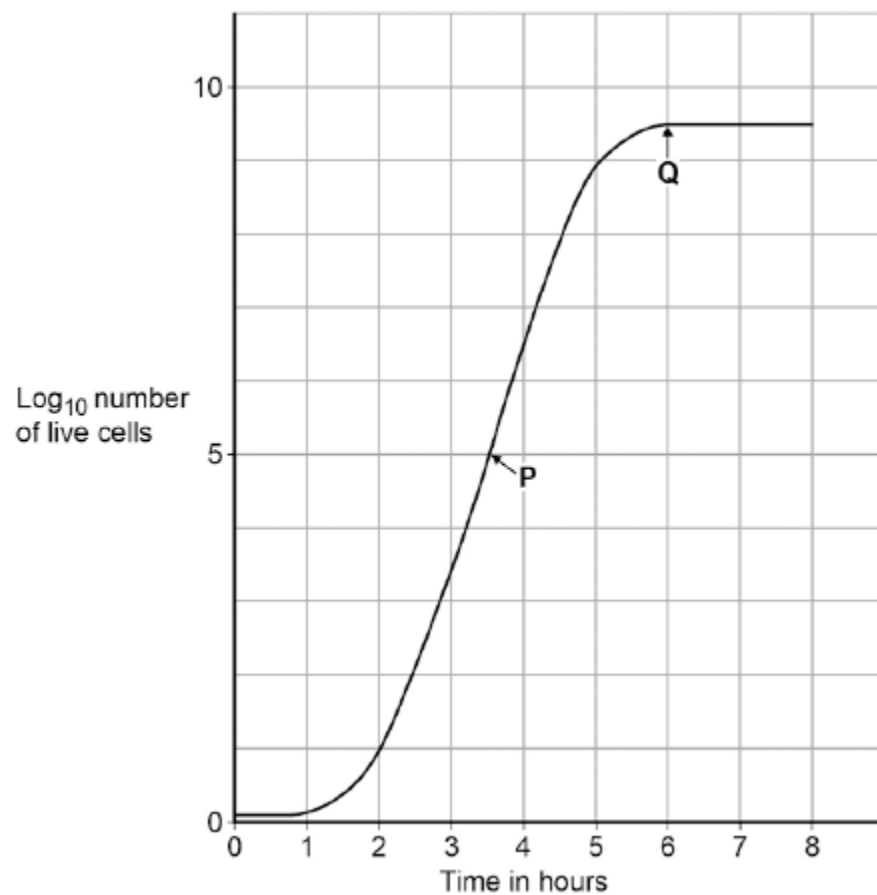
Time in **Stage 2** = _____ minutes

(3)

- (e) Bacteria such as *Escherichia coli* undergo cell division similar to mitosis.

Figure 2 shows a growth curve for *E. coli* grown in a nutrient broth.

Figure 2



What type of cell division causes the change in number of *E. coli* cells at **P**?

(1)

(f) Suggest why the number of cells levels out at Q.

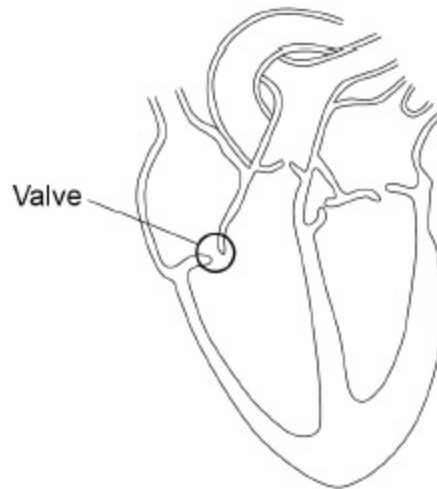
(2)

(Total 11 marks)

4.

The figure below shows the internal structure of the human heart.

One of the heart valves is labelled.



Sometimes a valve in the heart can start to leak.

(a) Explain why a person with a leaking heart valve has difficulty exercising.

(4)

A patient with a leaking heart valve may have the valve replaced.

A study compared two different types of replacement heart valve:

- mechanical valves
- biological valves from pigs.

The data used in the study was collected from female patients aged 50–69.

The following table shows the data.

	Type of replacement heart valve	
	Mechanical	Biological
Number of patients given the valve	2852	1754
Number of patients who died from heart-related problems after valve replacement	180	178
Percentage of patients alive after 5 years	91	89
Percentage of patients needing a second valve replacement within 6 years	2.2	5.2
Percentage of patients who had a blood clot on the brain after surgery	5.8	0.1

- (b) Give **one** conclusion about the death of patients from heart-related problems after a valve replacement.

Include calculations to support your answer.

(3)

- (c) One risk of mechanical valves is that blood clots can form on the surface of the valve.

Name the component of the blood that starts the process of blood clotting.

(1)

- (d) Evaluate the use of mechanical replacement heart valves and biological replacement heart valves.

Use information from the table above and your own knowledge.

(6)

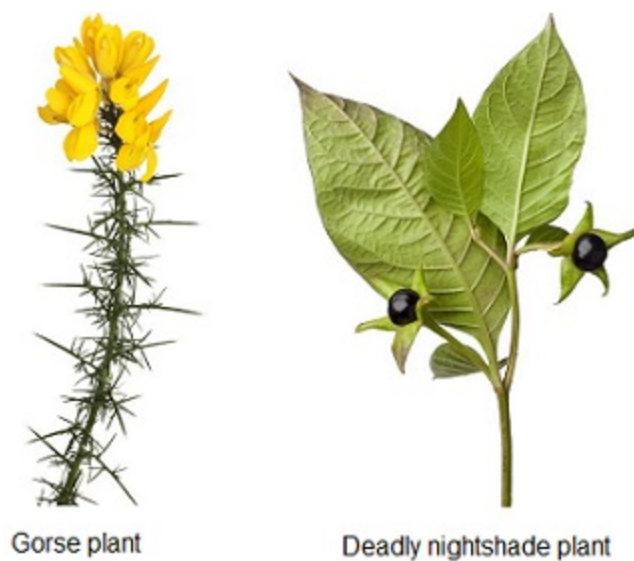
(Total 14 marks)

5.

Many plants have evolved defence mechanisms.

Figure 1 shows part of a gorse plant and part of a deadly nightshade plant.

Figure 1



- (a) The gorse plant has evolved to have sharp thorns.

What type of defence response are thorns?

(1)

- (b) How do thorns defend the gorse plant?

(1)

- (c) The deadly nightshade plant has poisonous berries.

What type of defence response are poisonous berries?

(1)

- (d) A scientist noticed that in one area the gorse plants had yellow leaves and had stunted growth.

One reason for yellow leaves and stunted growth is a deficiency of nitrate ions in the soil.

Explain **two** other possible reasons for the yellow leaves and stunted growth.

Do **not** refer to nitrate ions in your answer.

Reason 1

Explanation

Reason 2

Explanation

(5)

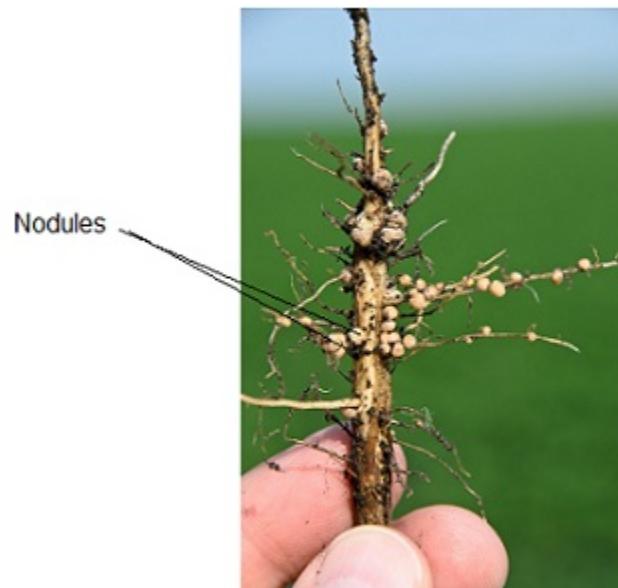
The gorse plant has nodules on its roots.

The nodules are part of the living root tissue.

Bacteria which convert nitrogen gas into soluble nitrate ions live in the nodule tissue.

Figure 2 shows the nodules on the roots.

Figure 2



- (e) Suggest how the nodules benefit the bacteria.

(2)

- (f) Explain how the nodules benefit the gorse plant.

(2)

- (g) For many years drugs have been extracted from plants.

Which plant material was chewed as a painkiller?

Tick (✓) **one** box.

Blackcurrant berries

☐

Foxglove leaves

☐

Rose petals

☐

Willow bark

☐

(1)

(Total 13 marks)

Mark schemes

1.

- (a) **Level 2:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

3–4

Level 1: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.

1–2

No relevant content

0

Indicative content

- use of quadrat
- (quadrat) of given area / dimensions – e.g. 0.25 m² or 1 m × 1 m
- quadrats are placed randomly
- method of obtaining randomness – e.g. random coordinates from a calculator **or** throw over shoulder **or** throw with eyes closed
- suitable number of quadrats (10 or more **or** a large number)
- count number of plants (in each quadrat)
- calculation of mean per quadrat or per unit area
- determination of area of field (length × width)
- population = mean per m² × area of field

- (b) more bacteria so more divisions / reproduction (per unit time)

1

- (c) any **three** from:

- add (more) sugar
- add (more) amino acids / protein
 - if neither point given, allow add (more) nutrients*
- add (more) oxygen
- increase temperature
 - allow in range 26 °C to 40 °C*
 - allow maintain optimum temperature*
- remove toxins / waste **or** maintain pH
- stir the culture
 - if no other mark awarded allow 1 mark for add more food*

3

(d)

an answer in the range of 2.9 to 3.4 scores 4 marks

an answer in the range of 2.08 to 3.77 scores 3 marks

tangent drawn to the curve at 12 hours

*do **not** accept if there is an incorrect tangent at 7 hours*

1

calculation of rate at 7 hours $\frac{\Delta y}{\Delta x}$

allow an answer that correctly rounds to a value in range 10.0 to 11.7

1

calculation of rate at 12 hours $\frac{\Delta y}{\Delta x}$

allow an answer that correctly rounds to a value in range 3.1 to 4.8

1

3.3

allow in range 2.9 to 3.4 if both rates are in the correct ranges

1

(e) can use the glyphosate / weed killer to kill weeds but not kill / affect crop

*allow **only** kills weeds*

1

(so) less competition for light / water / minerals / ions

allow less competition for nutrients

ignore food / carbon dioxide / space

1

(so) crops have high(er) yield

allow crops grow better / well

1

[15]

2.

Level 3 (5–6 marks):

A detailed and coherent explanation is provided with most of the relevant content, which demonstrates a comprehensive understanding of the human circulatory system . The response makes logical links between content points.

Level 2 (3–4 marks):

The response is mostly relevant and with some logical explanation. Gives a broad understanding of the human circulatory system. The response makes some logical links between the content points.

Level 1 (1–2 marks):

Simple descriptions are made of the roles of some of the following: heart function, gas exchange, named blood vessels, named blood cells. The response demonstrates limited logical linking of points.

0 marks:

No relevant content.

Indicative content

- dual / double circulatory system which means that it has higher blood pressure and a greater flow of blood to the tissues
- heart made of specialised (cardiac) muscle cells which have long protein filaments that can slide past each other to shorten the cell to bring about contraction for pumping blood
- heart pumps blood to lungs in pulmonary artery so that oxygen can diffuse into blood from air in alveoli
- blood returns to heart via pulmonary vein where muscles pump blood to the body via aorta
- oxygen carried by specialised cells / RBCs which contain haemoglobin to bind oxygen and have no nucleus so there is more space available to carry oxygen
- arteries carry oxygenated blood to tissues where capillaries deliver oxygen to cells for respiration and energy release
- thin walls allow for easy diffusion to cells
- large surface area of capillaries to maximise exchange
- waste products removed eg CO₂ diffuse from cells into the blood plasma
- blood goes back to the heart in veins which have valves to prevent backflow
- cardiac output can vary according to demand / is affected by adrenaline

accept annotated diagrams

[6]

3.

(a) C

1

(b) cytoplasm **and** cell membrane dividing
accept cytokinesis for 1 mark

1

to form two identical daughter cells

1

(c) stage 4

1

only one cell seen in this stage

1

(d) $(4 / 36) \times 16 \times 60$

1

107 / 106.7

1

110 (minutes)

allow 110 (minutes) with no working shown for 3 marks

1

(e) binary fission

*do **not** accept mitosis*

1

(f) shortage of nutrients / oxygen

1

so cells die

or

death rate = rate of cell division

1

[11]

4.

(a) **Level 2:** Relevant points (reasons/causes) are identified, given in detail and logically linked to form a clear account.

3–4

Level 1: Relevant points (reasons/causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.

1–2

No relevant content

0

Indicative content:

- backflow can occur **or** some blood flows backwards
- less blood leaves the heart **or** less blood is pumped around the body **or** some blood stays in the heart (instead of being pumped out) **or** reduced blood pressure **or** reduced flow rate
- less oxygen supplied to muscles / cells
- (so) less aerobic respiration
- (so) less energy released
- (so) less (efficient) muscle contraction
- anaerobic respiration takes place
- less (efficient) removal of lactic acid **or** lactic acid builds up **or** oxygen debt occurs
- (lactic acid building up) causes muscle fatigue
- less (efficient) removal of carbon dioxide (from blood)

a **level 2** response should refer to both respiration **and** the effects on exercise

(b)

ignore raw numbers from the table

(deaths mechanical valve =) 6% / 6.31136%

allow correctly rounded value

1

(deaths biological valve =) 10% / 10.14823%

allow correctly rounded value

1

(therefore a) higher proportion / percentage of patients die with biological valve

or

patients are more likely to die with biological valve

*do **not** accept more patients die with a biological valve*

*allow **2** marks for ratio mechanical : biological = 1:1.6 **or***

*1:1.7 **or** correctly calculated value*

*allow **3** marks for deaths with biological*

*valves = 4% / 3.83687% higher **or** correctly rounded value*

or

patients are 1.6 / 1.7 times more likely to die with biological valves

*if **no** other marks awarded, allow for **1** mark*

*chance of death after a valve replacement is 8% / 7.77247% **or** correctly rounded value*

1

(c) platelets

allow thrombocytes

1

(d) **Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.

5-6

Level 2: Some logically linked reasons are given. There may also be a simple judgement.

3-4

Level 1: Relevant points are made. They are not logically linked.

1-2

No relevant content

0

Indicative content:

mechanical valves

- longer lasting **or** more durable **or** don't wear out as easily **or** less likely to need replacing (within 6 years)
- blood clots (on the brain) are more likely (after surgery)
- patient has to take anti-clotting medication (for the rest of their lives)
- if medication not taken (correctly), clots can lead to blood clots on brain / heart attack
- medication can lead to excessive bleeding (after injury)
- some patients say they can hear the valves opening and closing
- survival rate at 5 years is slightly higher for mechanical valve
- lower percentage of deaths due to heart-related problems

biological valves

- no additional medication required
- ethical issues surrounding use of animal tissue
- valve may harden
- more likely to need further operation **or** another new valve
- more likely to be rejected
- more likely to need (immuno-suppressant) medication

both valves

- both are readily available
- little wait time

a **level 2** response should contain comparisons of both valves **and** some reference to own knowledge

[14]

5.

(a) mechanical

allow physical
allow structural

1

(b) any **one** from:

- to deter herbivores
ignore to injure animals, unqualified
allow to deter animals eating it
*do **not** accept to deter predators*
- to prevent animals damaging it

1

(c) chemical

1

(d) any two from :	5
• lack of magnesium (ions) (1)	1
(so) not enough chlorophyll for (efficient) photosynthesis (1)	
(so) not enough glucose to make proteins for growth	
or not enough glucose to release energy for growth (1)	
<i>allow (so) lack of chlorophyll produced causes yellow leaves (1), (so) not enough photosynthesis to produce glucose which is used to make proteins for growth (1)</i>	
• infection by pathogen / bacteria / virus / fungus (1)	
<i>allow correctly named pathogen</i>	
<i>allow has rose black spot / TMV</i>	
(so) leaves become discoloured / yellow so less photosynthesis (1)	
<i>allow other symptoms of named pathogens / disease</i>	
(so) not enough glucose to make proteins for growth or not enough glucose to release energy for growth (1)	
<i>award once only</i>	
• infected by aphids (1)	
(which) remove sugars from phloem (1)	
(so) not enough glucose to make proteins for growth or not enough glucose to release energy for growth (1)	
<i>award once only</i>	
• lack of (available) light (1)	
(so) chlorophyll breaks down (1)	
(so) not enough glucose to make proteins for growth or not enough glucose to release energy for growth (1)	
<i>award once only</i>	
(e) (bacteria) obtain glucose / sugar (from the plant)	1
(glucose used) for respiration or (glucose used) for making other named substances	
<i>allow (glucose used) to release energy</i>	1
(f) (gorse plant) obtains nitrate (ions)	1
needed for amino acids / proteins	
<i>allow needed to make chlorophyll / DNA</i>	

(g) willow bark

1
[13]

Examiner reports

1.

- (a) This extended response question related to RPA 9: Measure the population size of a common species in a habitat. The species here was the weed, ragwort, growing in a field on a farm.

The mark scheme was divided into two levels and the requirement for entering Level 2 was that the method would lead to the production of a valid outcome with all key steps being identified and logically sequenced. In relation to the topic of the question, this meant that:

- a suitable number of quadrats should be placed (a minimum value of ten)
- a method for placing the quadrats randomly should be described
- the number of ragwort plants within the quadrat should be counted
- the mean value for unit area (say, 1 m²) should be scaled up to the area of the field.

This meant that the placing of 'several quadrats'/'a few quadrats'/'five quadrats' was insufficient to enter Level 2, as did 'throwing the quadrat randomly'. Ideally, each quadrat should have been placed at a pair of coordinates selected randomly from a table of random numbers or using the random number button on a calculator or even by pulling numbers randomly out of a hat. Examiners did allow throwing the quadrat if it was done 'with eyes shut' or 'over the shoulder' but such a practice is not to be recommended and is not considered to be a valid method of achieving randomness.

Some students decided to place quadrats at intervals along a transect line which was not really appropriate for estimating a population size; others determined percentage cover which, again, would not give a population size. Three-quarters of students did not give answers above Level 1.

- (b) This question asked for an explanation of why the rate of reproduction of the bacteria increased between two times on the graph (in the exponential phase of growth). A quarter of students were able to explain that since there were more bacteria with the passage of time, so there would be more bacteria to reproduce at the later time.
- (c) As in most typical growth curves, the rate of growth began to slow down after a certain time. Suggestions made by students for how a high bacterial reproduction rate could be maintained included:
- adding more nutrients (or named examples)
 - adding more oxygen
 - increasing the temperature
 - removing any toxic waste products
 - simply stirring the culture.

Most students were able to give at least one of these with around 10% of students able to give the three asked for.

- (d) Two rates of reproduction of the bacteria had to be compared in this question. The rate at 7 hours was quite straightforward as the graph was a straight line at this time. To determine the rate at 12 hours, students had to draw a tangent to the curve (specification WS 3.3).

Many students successfully determined the rate at each of the two specified times and then divided the rate at 7 hours by the rate at 12 hours. Since there was reasonable variation between tangents drawn by different students, examiners accepted answers for the rates and for the final quotient within certain ranges, as well as ascertaining that tangent(s) were drawn on the graph.

This question differentiated well between students of differing abilities, with approximately 11% gaining full marks and over half gaining partial credit. The greatest success was shown by those who drew their tangents over the full extent of the graph paper – thus any small errors in reading the scales had a smaller proportional effect.

Some students did not draw tangents but divided the number of bacteria at 7 hours by 7 and that at 12 hours by 12 to find the two rates and, undeterred by the outcome of the rate at 12 hours appearing to be greater than the rate at 7 hours, simply divided the larger number by the smaller.

- (e) The most obvious advantage of genetic modification of crop plants to make them resistant to a herbicide, was that weeds could be killed without harming the crop. Almost three quarters of students realised this and gained credit. Some students went on to explain that this would result in a higher yield of crop due to reduced competition with weeds for some named factor e.g. water, light or mineral ions. Around 7% of students gained full marks.

4.

- (a) 11% of students gave a response in line with the Level 2 descriptor. Many students were unable to logically link their ideas about increased backflow of blood to the effect on exercise. This, and a reference to respiration, was required for Level 2. Students who struggled to gain any credit in this question simply stated the function of the valves within the heart.

Many students realised that a faulty valve would allow backflow of blood, and a high proportion of those continued on to say that this would lead to problems with overall blood circulation. Slightly better responses explained that this would lead to a lack of oxygen to the cells/tissues, and so aerobic respiration would be limited, or anaerobic respiration would occur. The majority of students, if they reached this point, went no further. Very few students then completed the account to relate these points to exercise, and how muscles were less able to function.

Whilst some students were able to make reference to decreased respiration, very few described the type of respiration they were referring to (aerobic or anaerobic) or how this decrease affected exercise. The students who did refer to anaerobic respiration were more likely to achieve Level 2 than those referring to aerobic respiration, as many often went on to refer to muscle fatigue. Many came close with a reference to lactic acid but failed to link this to its effect on muscles. References to fatigue in general were commonly seen, but this was not specific enough to be creditworthy at this level.

A significant misunderstanding which manifested itself regularly in various parts of the paper and is repeated every year is that respiration 'produces', 'makes' or 'generates' energy. Students should be advised during their studies that this is not only unacceptable but is scientifically impossible. A further common misconception seen was that a leaking heart valve would lead to the mixing of oxygenated and deoxygenated blood, rather than a backflow of (some) blood.

- (b) The success rate for correct calculations in this question was high, and many students drew the correct conclusion from their result.

A considerable number of students however, failed to read the question correctly and gave the wrong figures, quoting the percentages of those alive after 5 years or those developing blood clots. Common errors included students failing to manipulate the data, and simply quoting raw figures, leading of course, to the wrong conclusion, that biological valves are better.

Students should be encouraged to check their calculations for rounding errors, as these occurred frequently.

- (c) 87% of students were able to identify platelets as the blood components that start the clotting process, however, the spelling of the terms was often incorrect.
- (d) It was clear where students had clearly identified the word 'evaluate' in the stem of the question as their responses showed good use of comparative statements.

Weaker responses indicated that students had failed to read the question carefully enough or did not understand the word evaluate. A large number of these responses didn't include comparisons and were just facts about one type of valve. In some cases where comparisons were given, the wording of the sentence made it unclear as to which type of valve was being referred to and students should therefore be encouraged to read back over their responses to check for clarity, if they have time.

Many students included the calculations they had carried out in part (b) to support their comparisons. Some students attempted to work out further percentages using figures from the table that were already given as percentages. Students who failed to notice that the data for death from heart related problems had not been converted to a percentage invariably used the higher number of deaths after mechanical valve replacement as a negative factor. There were also a lot of responses that included incorrectly calculated figures which led to incorrect conclusions. Many students simply misquoted the figures given to them or gave the comparisons in favour of the wrong type of valve.

The biggest hurdle that prevented students scoring well on this question was the instruction to use their own knowledge. Without reference to any scientific detail not included in the table, the student's answer was limited to Level 1 (maximum two marks). The majority of students gave many comparisons of the information supplied to them but were still restricted to the lowest level. A Level 3 answer required a judgement, either stated or implied, supported by comparisons of the data and own knowledge.

The most commonly seen own knowledge referred to was the idea of rejection. This was accepted for pig valves even though in reality, they are treated so that they are not rejected. As the source is living tissue, it was accepted that many students would assume the pig valves contain antigens that the recipient's body will react to. Just the mention of rejection was enough to elevate a student's answer into Level 2 but was considered weak own knowledge unless supported by some more detail, such as the use of immunosuppressant drugs. Some students got the idea of rejection the wrong way round, stating that the mechanical valves would be more likely to be rejected, which was not credited.

Some students were able to link the risk of blood clots in the brain with strokes or heart attacks, and the need for anti-clotting drugs, although some talked about the need for the medication without giving a reason why.

Only a minority of students mentioned the ethical or religious issues surrounding the use of pig tissue and very few mentioned the hardening of the valves or the fact that mechanical valves can sometimes be heard working.

Very few students included any of the indicative content listed about both valves. In fact, rather than realising that both valves are readily available, several students referred to the need to wait for a donor, with the implication that the valves were coming from human donors rather than pigs. This idea was not credited.

Other ideas not deemed creditworthy were those of cost or infections due to surgery. Both of these could apply to either type of valve so were not credited as a comparison.

The presentation of many student responses caused issues, as many wrote a long introduction to their answer, which was unnecessary, and this caused them to run out of space and continue their answers all over the page. Whilst most students should not need to use additional answer space, students should be reminded to use an additional answer booklet, rather than trying to squeeze answers in the margin or at the bottom of the page. Many answers showed students' lack of awareness of how to tackle an extended writing answer - some of them answered very briefly with one or two simple statements and others were written in general terms without reference to the data provided.

5.

Some students referred incorrectly to herbivores as predators in (b). In (d), many students were able to correctly name two reasons for the symptoms, but explanations were either not worded well or underdeveloped. Several students were unclear about the distinctions between chlorosis, chloroplasts and chlorophyll.

Very few students answered (e) correctly (89% scored zero marks), because they were so focussed on the nitrates mentioned in the question that they missed the other requirements of the bacteria. Many students found the novel context regarding root nodules perplexing and didn't score any marks.